Lesson 14: Experimental Design: Between and Within Subjects

English Stage 4 (Psychology) · Unit 3: Research Designs

Adapted from Research Methods in Psychology (4th American edition) by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler and Dana C. Leighton (CC BY-NC-SA 4.0): 5.3: Experimental Design.

In this lesson

You will compare the two basic ways of designing an experiment: testing each participant in one condition (between subjects) or in all conditions (within subjects). You will see how random assignment and counterbalancing stop extraneous variables from becoming confounds, and practise the language of comparison.

Learning goals

After this lesson you should be able to:

- Explain the difference between between-subjects and within-subjects experiments and list the pros and cons of each.
- Distinguish random assignment from random sampling, and describe simple and block randomization.
- Define order, carryover, practice, fatigue and context effects, and explain how counterbalancing deals with them.
- Choose and justify a design, comparing the two options in clear academic sentences.

Key language

Term or phrase	Meaning	Example
between-subjects experiment	an experiment in which each participant is tested in one condition	Half write about a traumatic event, half about a neutral event.
within-subjects experiment	an experiment in which each participant is tested in all conditions	The same participants judge an attractive and an unattractive defendant.
random assignment	assigning participants to conditions by a random procedure	flipping a coin for each participant
random sampling	selecting a sample from a population at random (rare in psychology)	Do not confuse random assignment with random sampling.
block randomization	random assignment in which all conditions occur once, in random order, before any is repeated	A C B / B C A / C B A

Term or phrase	Meaning	Example
matched-groups design	participants are matched on the DV or an extraneous variable before being assigned to conditions	pairs of equally healthy participants
carryover effect	an effect of being tested in one condition on behaviour in later conditions (one kind of order effect)	judging more harshly because of boredom
practice effect / fatigue effect	better / worse performance in later conditions because of practice / tiredness or boredom	doing better on the second attempt
context effect (contrast effect)	an earlier condition changes how participants perceive stimuli or interpret their task	An average-looking defendant seems worse after an attractive one.
counterbalancing	systematically varying the order of conditions across participants	Half do A then B; half do B then A.
the former ... the latter	the first ... the second of two things just mentioned	The former are called between-subjects experiments and the latter are called within-subjects experiments.
the primary advantage / disadvantage of X is that ...	introduces the most important strength / weakness	The primary disadvantage of within-subjects designs is that they can result in order effects.

Rules and explanations

The ideas

Between-subjects experiments and random assignment

In a **between-subjects experiment**, each participant experiences one level of the IV (for example, 60 people with severe agoraphobia, 20 in each of three treatments). The groups must be highly similar on average (sex, IQ, motivation, health), so that these participant variables do not become confounds.

- **Random assignment**, strictly, meets two criteria: each participant has an equal chance of each condition, and each is assigned independently (a coin flip for two conditions; a random integer from 1 to 3 for three).
- **Block randomization** avoids unequal group sizes (equal groups are statistically most efficient): within each block every condition occurs once, in random order. The book's table assigns nine participants as A C B, B C A, C B A. Never throw away collected data to make groups equal.
- Random assignment can fail by chance, but for three reasons this is not a major concern (Reading question 4). It is always considered a strength of a design.

- **Matched groups:** rank participants on a variable (for example, health), then randomly assign each pair of similar participants to different conditions, so that this variable cannot be confounded with the IV.

Within-subjects experiments and counterbalancing

In a **within-subjects experiment**, each participant is tested in all conditions. **Primary advantage:** maximum control of participant variables (the same people, so the same IQ, social class and so on), less noise, and fewer participants needed. **Primary disadvantage:** order effects, when responses are affected by the order of conditions.

- **Carryover effects:** a practice effect, a fatigue effect, or a context effect.
- Participants may also guess the hypothesis and respond as they think they are expected to, or try to be "fair".
- If the attractive defendant always comes first, order becomes a confounding variable.

Counterbalancing tests different participants in different orders, assigned randomly. "If a study does not involve random assignment in one form or another, it is not an experiment."

Method	How it works	Orders needed
Complete counterbalancing	every possible order, equally often	3 conditions: 6 (ABC, ACB, BAC, BCA, CAB, CBA); 4: 24; 5: 120; 6: 720
Latin square	each condition appears once in each position	as many orders as conditions (6 for 6 conditions)
Random counterbalancing	a random order for each participant	for many conditions; adds random error

Counterbalancing does two things: it stops order from being a confound, and it makes carryover effects detectable (analyse each order separately).

A caution on the Latin square. The book says that in its square "each condition precedes and follows each other condition one time". Check the square: A always comes immediately before B (rows 1, 3 and 4), and B never comes immediately before A. The first claim (each condition once in each position) is true; the second is true only of a *balanced* Latin square, for example ABDC, BCAD, CDBA, DACB.

Box "When 9 Is 'Larger' Than 221": in Birnbaum's (1999) between-subjects study, the number 9 received a mean rating of 5.13 and 221 a mean of 3.10 on a 1-10 scale of size. Participants compared 9 with other one-digit numbers and 221 with three-digit numbers. Birnbaum argues that the lack of context in between-subjects designs is often a bigger problem than the context effects of within-subjects designs.

Simultaneous within-subjects designs mix the conditions in one sequence, for example 10 attractive and 10 unattractive defendants in mixed order.

Choosing a design

	Between subjects	Within subjects
Advantages	conceptually simpler; less testing time per participant; no carryover effects	controls participant variables; less noise; fewer participants
Disadvantages	more noise from participant differences; more participants	order and carryover effects; participants may guess the hypothesis

Rule of thumb: if a within-subjects design (with counterbalancing) fits the time available and carryover is not a serious concern, it is probably best. Choose between subjects when time per participant is short (a doctor's waiting room) or when a treatment produces long-term change: after a prejudice-reduction treatment works, participants are no longer suitable for the control condition. Researchers can also use both designs in different studies.

Language focus

Comparing and contrasting. The section compares two designs throughout. Useful patterns:

- *The former ... the latter:* "The former are called between-subjects experiments and the latter are called within-subjects experiments."
- *Advantage / disadvantage:* "Between-subjects experiments **have the advantage of being** conceptually simpler ..."; "The **primary disadvantage** of within-subjects designs **is that** they can result in order effects."
- *Comparative structures:* "Within-subjects experiments also require **fewer** participants **than** between-subjects experiments"; "This is **not as powerful a** technique **as** complete counterbalancing."

Note: *have the advantage of* + *-ing*; *not as* + adjective + *a* + noun + *as*.

> **Note for Arabic speakers** - After a preposition English needs the *-ing* form: "the advantage of **being** simpler", not "the advantage of be simpler".

Practice

1. Reading questions

1. How does random assignment differ from random sampling? Which is rarely used in psychological research?
2. What two criteria does random assignment meet in its strictest sense?
3. Why is block randomization standard practice? In the book's table, which conditions do participants 4, 5 and 6 receive?
4. Give three reasons why the fallibility of random assignment is not a major concern.
5. How would a matched-groups design work in the expressive writing study?
6. How many orders does complete counterbalancing need for four conditions? How many does a Latin square need for six?
7. What did Birnbaum (1999) find, and how did he explain it?

8. Why would a study that tries to reduce prejudice need a between-subjects design?

2. Book exercise

From the chapter exercises in Section 5.6 (printed with Lesson 16).

Discussion: For each of the following topics, list the pros and cons of a between-subjects and within-subjects design and decide which would be better.

- a. You want to test the relative effectiveness of two training programs for running a marathon.
- b. Using photographs of people as stimuli, you want to see if smiling people are perceived as more intelligent than people who are not smiling.
- c. In a field experiment, you want to see if the way a panhandler is dressed (neatly vs. sloppily) affects whether or not passersby give him any money.
- d. You want to see if concrete nouns (e.g., dog) are recalled better than abstract nouns (e.g., truth).

3. Language exercise

Join each pair into one sentence with the word(s) in brackets.

1. (whereas) In a between-subjects experiment, each participant is tested in one condition. In a within-subjects experiment, each participant is tested in all conditions.
2. (unlike) Random sampling is rarely used in psychological research. Random assignment is an important element of all experimental research.
3. (the former ... the latter) Complete counterbalancing of six conditions needs 720 orders. A Latin square needs only six.
4. (although) Within-subjects experiments control extraneous participant variables. They can produce carryover effects.
5. (have the advantage of) Between-subjects experiments require less testing time per participant.

Speak with your tutor

1. Explain random assignment and random sampling in your own words, and ask your tutor to correct your explanation and your English.
2. In PSYC 110 Lecture 17, Professor Bloom says there are "big individual differences in how much sleep people need". Discuss with your tutor what this means for a between-subjects or a within-subjects sleep experiment.
3. Ask your tutor to play a reviewer who believes within-subjects designs are always better. Defend the book's rule of thumb with the prejudice example.

Writing task

Research project: choosing between subjects or within subjects (Lessons 13-16 build your design and its validity).

Write a paragraph (200-250 words) that states and justifies the design of your sleep and memory experiment. The running example is between subjects (20 students randomly assigned to a nap or a

wake condition, the design of the practice data in Lessons 27-32), but you may argue for a within-subjects design. Include:

- the design you choose and at least two reasons from this lesson;
- one disadvantage of your choice and how you will reduce it;
- for a between-subjects design: how participants will be randomly assigned (for example, block randomization made with the Research Randomizer);
- for a within-subjects design: the orders of conditions, how participants will be assigned to them, and the likely carryover effects with your solution (for example, two equivalent word lists, counterbalanced, a week apart).

Language: compare the two designs (*whereas*, *unlike*, *the primary advantage of … is that*); use the future passive for procedures.

If you work on your own topic, write the same paragraph for your own experiment.

Self-editing checklist:

- The design is named correctly (between-subjects or within-subjects).
- At least two reasons come from the lesson.
- One disadvantage is admitted and dealt with.
- Random assignment, not random sampling, is described.
- Carryover effects are named precisely (practice, fatigue, context).
- Comparison structures are correct (*fewer … than*, *the advantage of + -ing*).

Paste your draft into the chat and ask your tutor for feedback. Then rewrite it.

Watch online

- Research Randomizer, the free tool the book recommends for block randomization sequences: https://www.randomizer.org
- PSYC 110 Lecture 17 (sleep, and individual differences in how much sleep people need): https://oyc.yale.edu/psychology/psyc-110/lecture-17

Answer key

1. Reading questions

1. Random sampling selects a sample from a population; random assignment assigns the participants in a sample to conditions. Random sampling is rarely used in psychology ("Random Assignment", paragraph 1).
2. Each participant has an equal chance of being assigned to each condition, and each is assigned independently of the others (paragraph 2).
3. Strict methods such as coin flipping often give unequal group sizes, and equal groups are statistically most efficient. Participants 4, 5 and 6 receive B, C and A: each block contains every condition once (paragraph 3 and table).
4. It works better than expected, especially with large samples; inferential statistics take its fallibility into account; a resulting confound would probably be detected when the experiment is replicated (paragraph 4).
5. Measure participants' health, rank them, then randomly assign the two healthiest to different writing conditions, then the next two, and so on ("Matched Groups").
6. 24 orders; a Latin square needs 6 ("Carryover Effects and Counterbalancing").
7. The number 9 got a mean rating of 5.13 and 221 a mean of 3.10. Participants compared 9 with one-digit numbers and 221 with three-digit numbers (box).
8. If the treatment works, participants' prejudice is reduced, so they are no longer suitable for testing in the control condition ("Between-Subjects or Within-Subjects?").

2. Book exercise

(example answers; other well-argued choices are possible)

- a. Between: no carryover, but more participants and more noise from fitness differences. Within: controls fitness differences, but the first program changes runners' fitness for months (a long-term change, like the prejudice example). **Better: between subjects**, perhaps with groups matched on fitness.
- b. Within: controls individual rating tendencies and needs fewer participants, but participants may guess the hypothesis, and context effects are possible. Between: avoids this, but needs more participants. **Better: within subjects**, with smiling and non-smiling photos in a mixed, counterbalanced order (a simultaneous within-subjects design).
- c. Within: passersby cannot be tested twice in a field setting and would notice a change of clothes. Between: simple, little time per person, no carryover. **Better: between subjects**, with the condition for each time period chosen by block randomization.
- d. Within: controls differences in memory ability and needs fewer participants; order effects are reduced by one mixed list of concrete and abstract nouns (a simultaneous within-subjects design). Between: needs more participants and adds noise. **Better: within subjects.**

3. Language exercise

(example answers)

1. In a between-subjects experiment each participant is tested in one condition, whereas in a within-subjects experiment each participant is tested in all conditions.
2. Unlike random sampling, which is rarely used in psychological research, random assignment is an important element of all experimental research.
3. Complete counterbalancing and a Latin square can both be used with six conditions, but the former needs 720 orders and the latter only six.
4. Although within-subjects experiments control extraneous participant variables, they can produce carryover effects.
5. Between-subjects experiments have the advantage of requiring less testing time per participant.

My notes